FGCT5004_24
Motion Capture Technologies
Development Log

By Henry Taylor 2206046

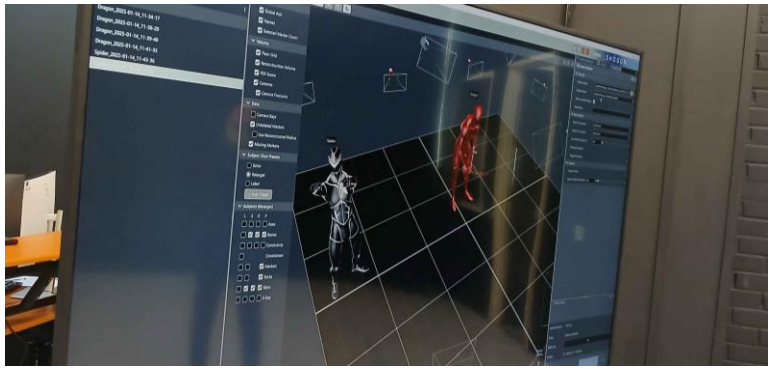
Getting suited and booted for the motion capture session

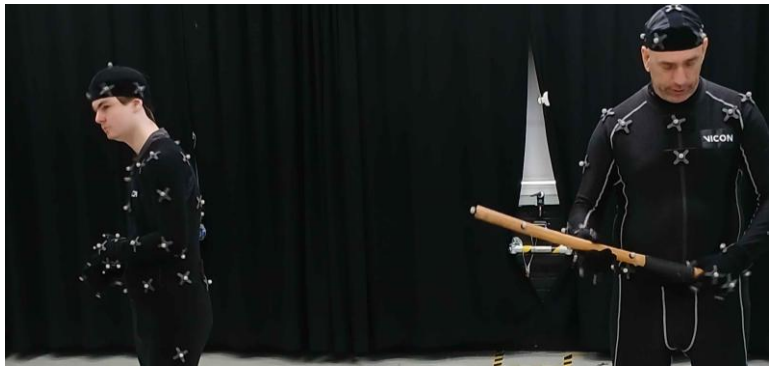












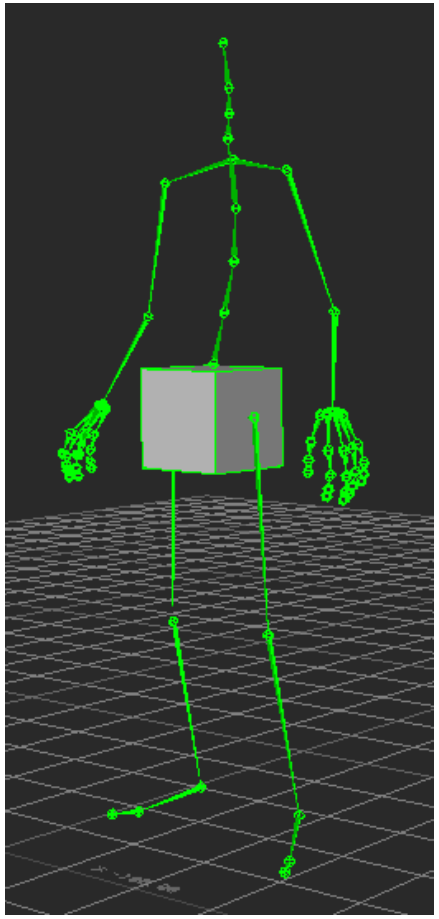
Some motion more capture action

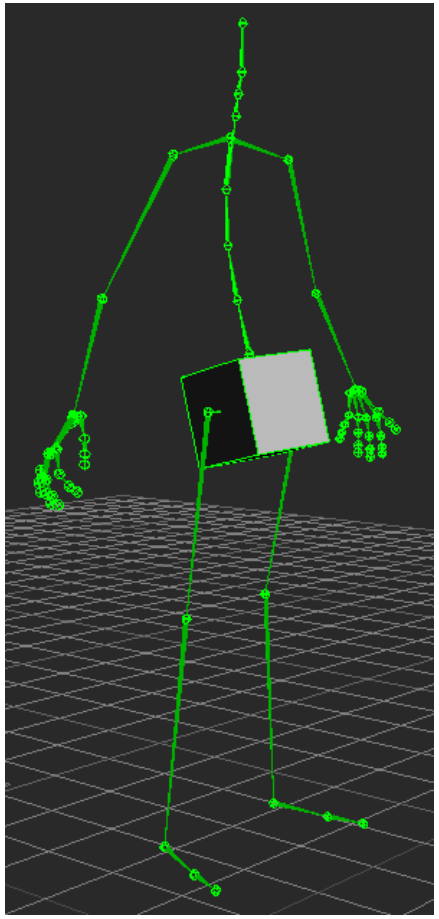


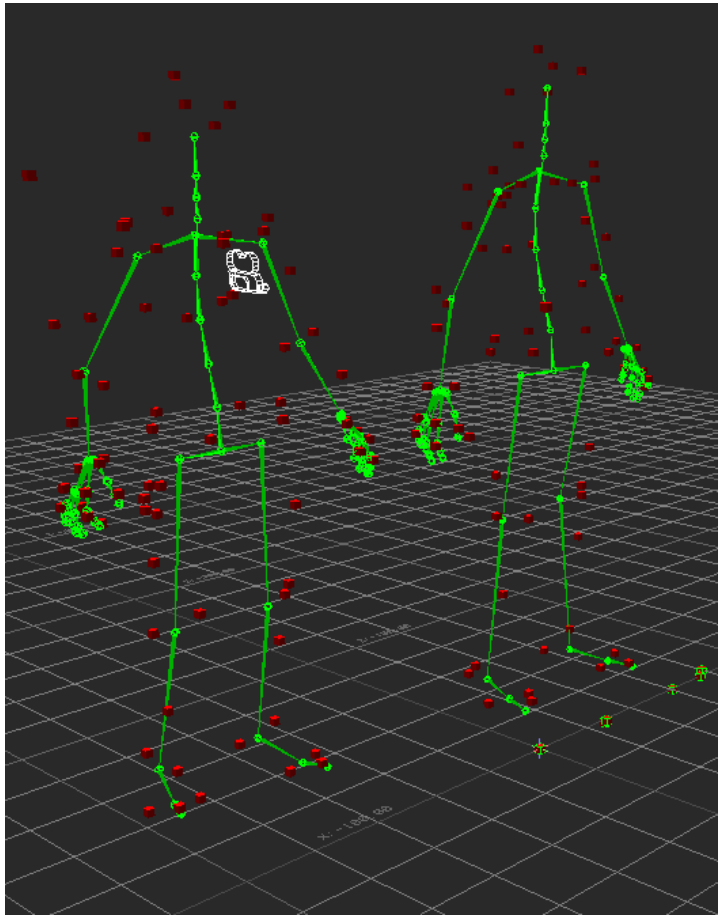


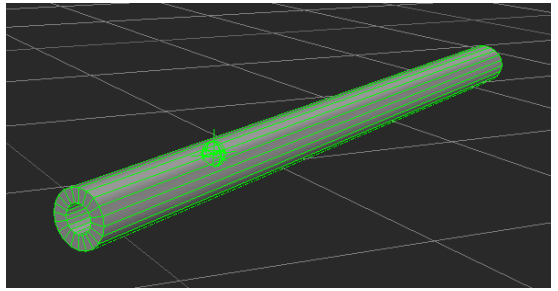


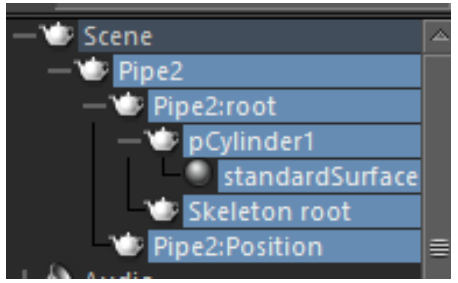

The facial capture and voice talent. Using iPhones with the LiveLink Face app.
On the left: David Awoliyi as Spider.
On the right: Orlando Cruz-Riquet as Dragon

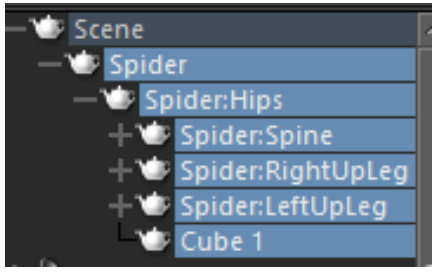


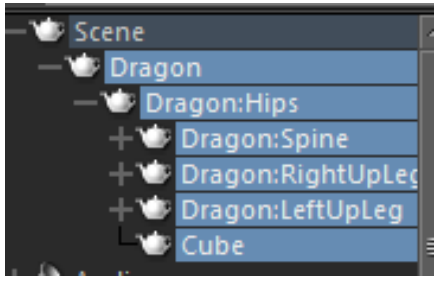










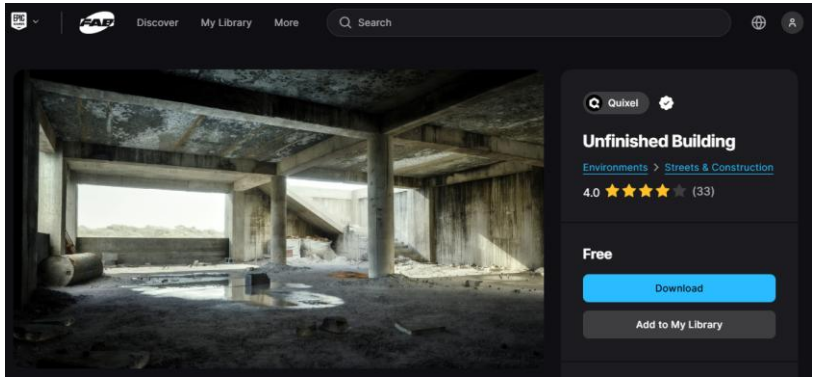


Setting up the roots and skeletons in Motion Builder in preparation for the animations.



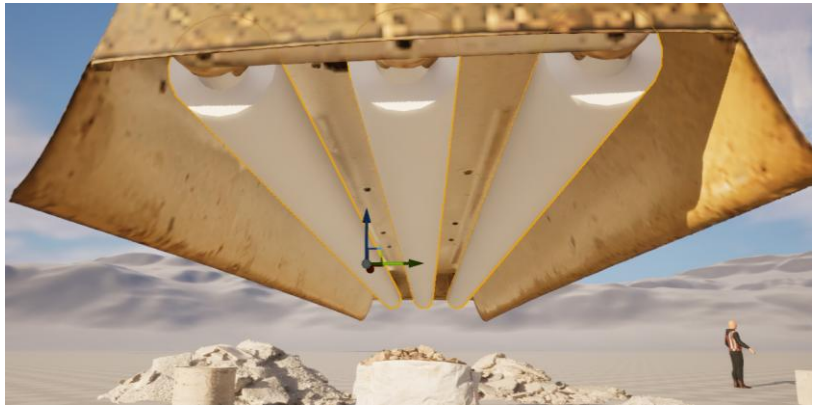
My two custom Meta-Humans. Spider on the Left and Dragon on the Right. I also adjusted their height as best as possible to match the height difference of the body motion capture performers.
Spider: Alex Tremlett
Dragon: Henry Taylor (me)

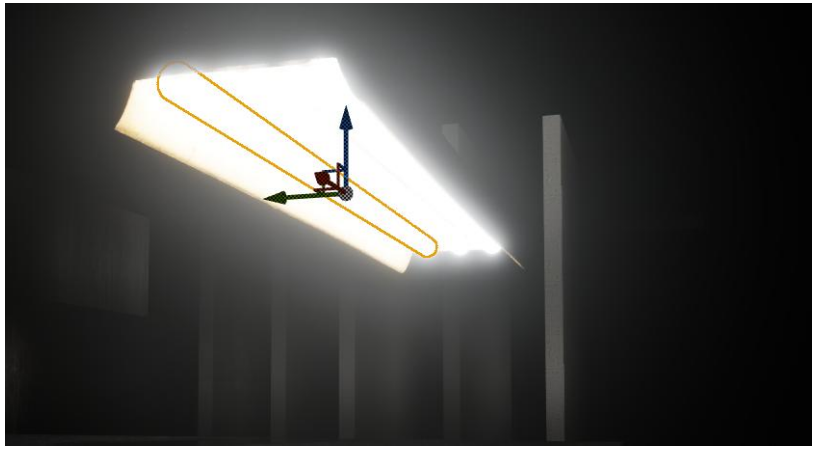




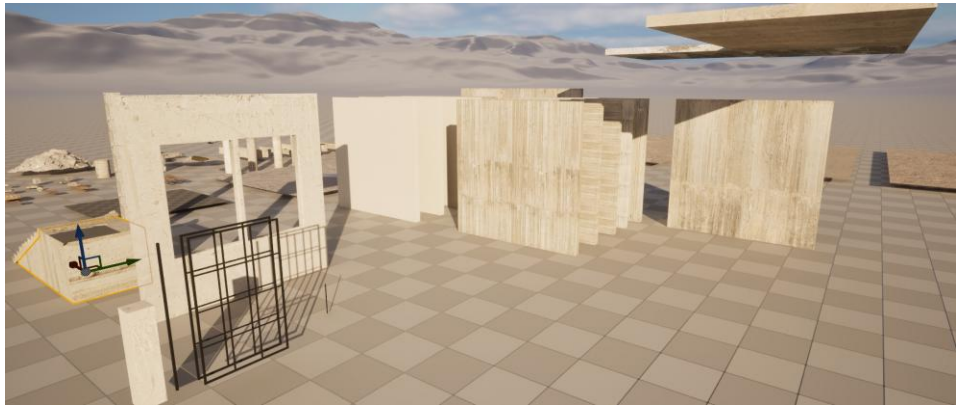




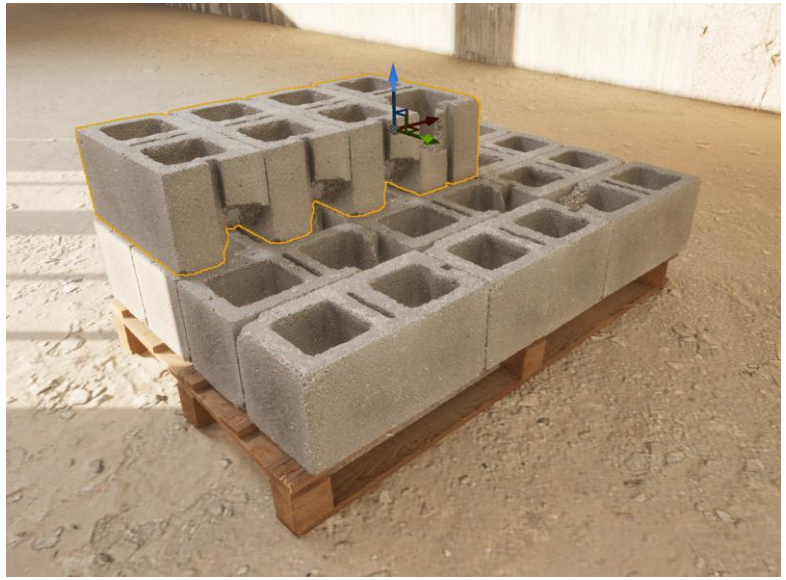


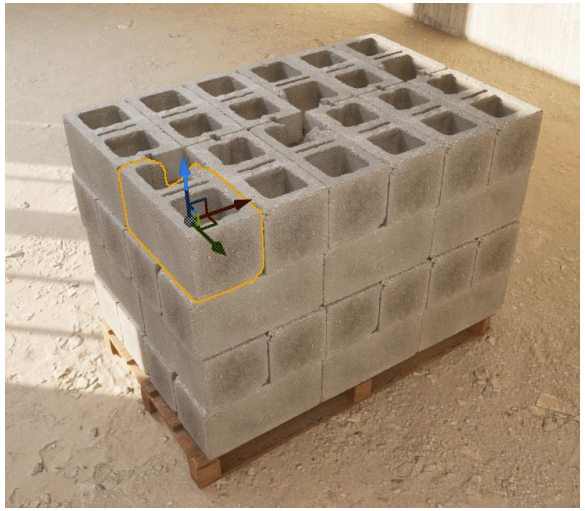


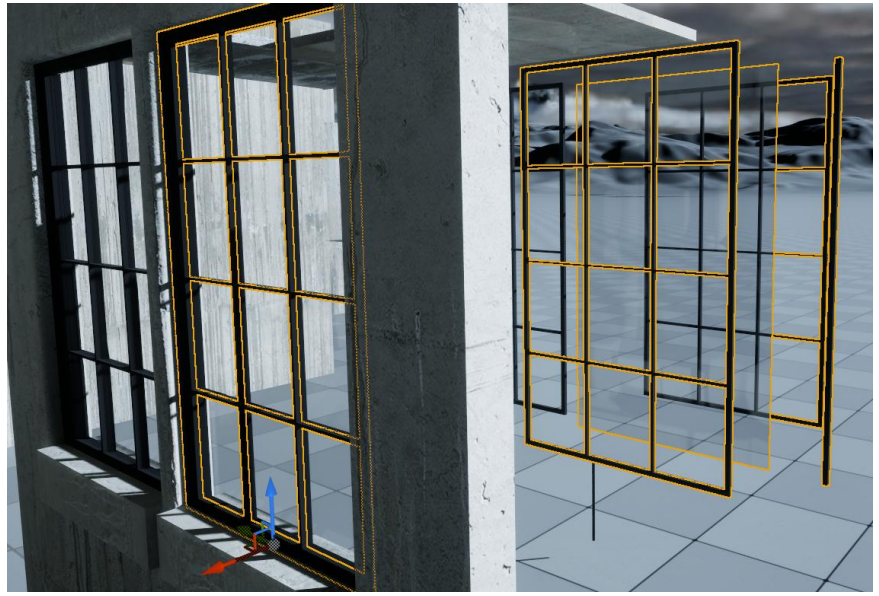
I used the Unfinished Building asset pack from UE5 Megascans Library for the environment.
I added an emissive material to the tube lights and also added a Light Source in engine to each of the placed lights.

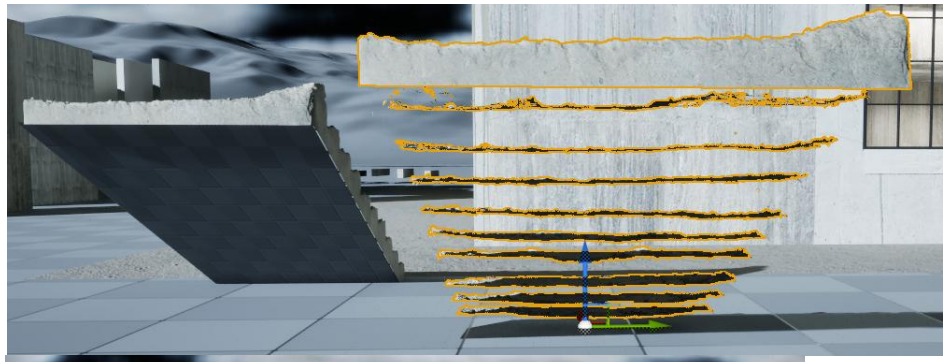






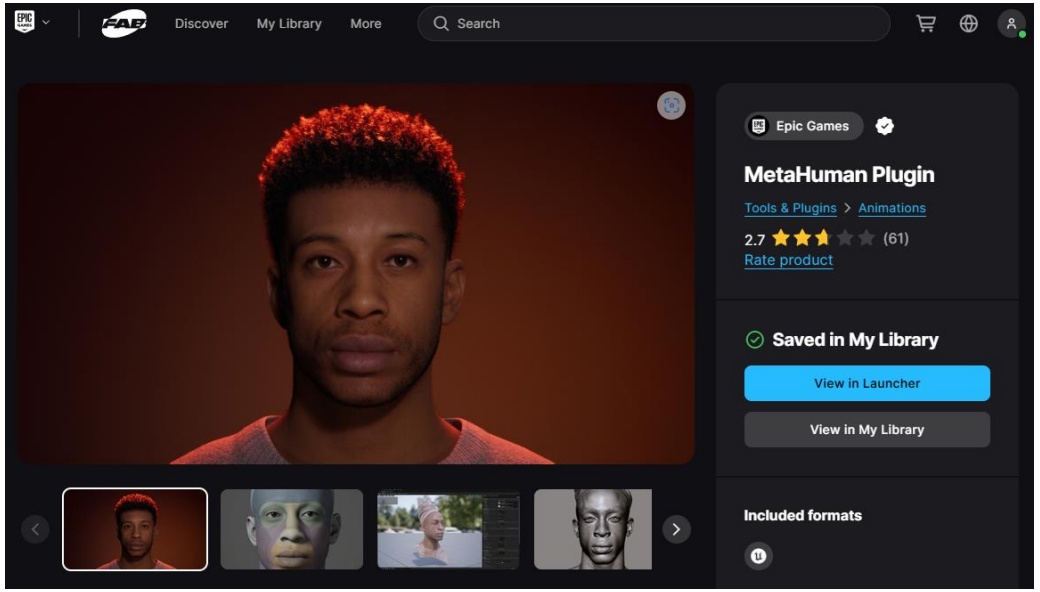


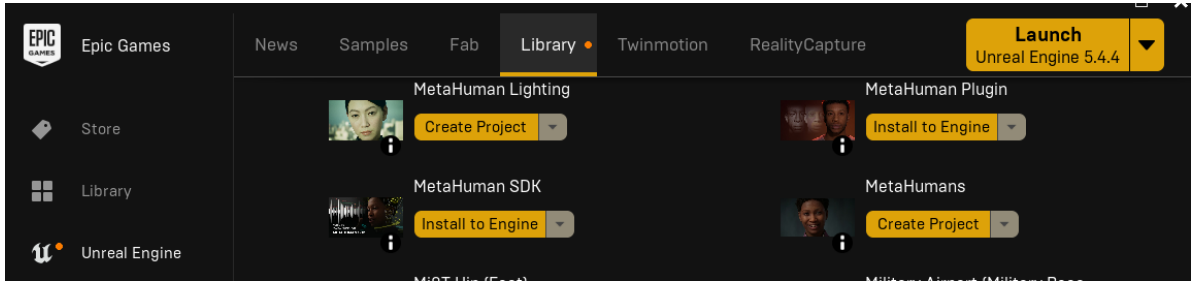




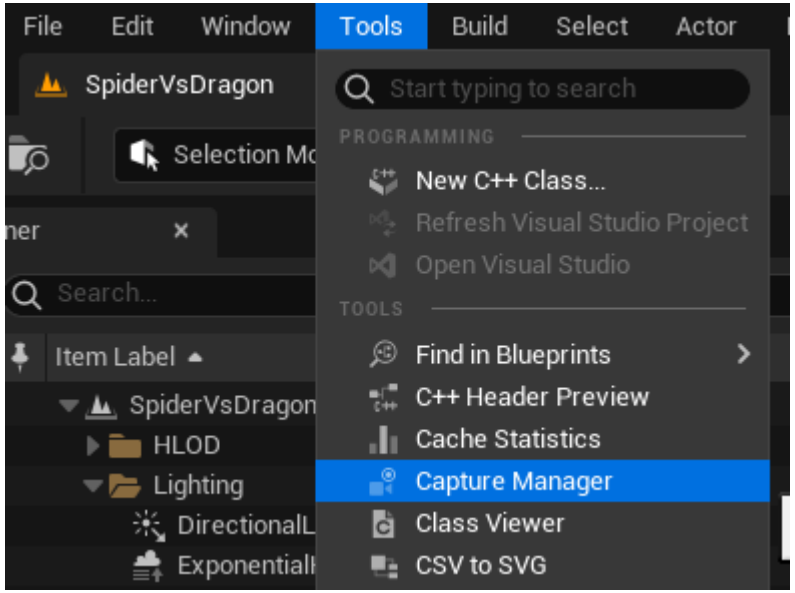


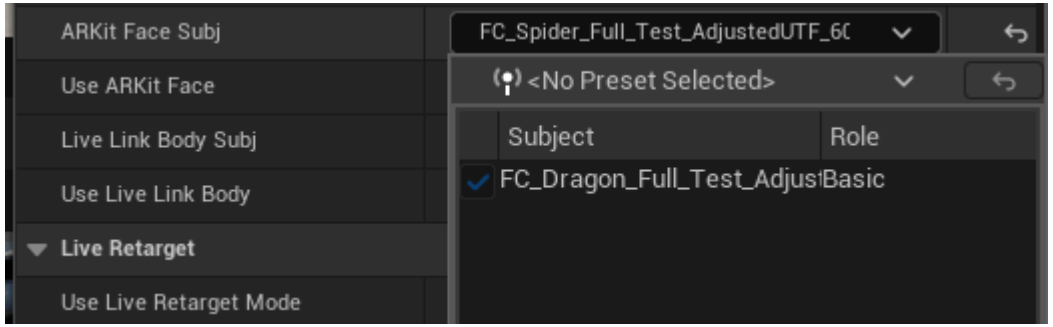
I made a pallet of breeze-blocks with the damaged ones that were in the asset pack.
I assembled the window from its constituent parts and duplicated it where needed in the environment.
I had to use a single plane to cover the under-side of the stairs as it would appear transparent when viewed from the backside and textured it using some of the concrete textures in the asset pack.



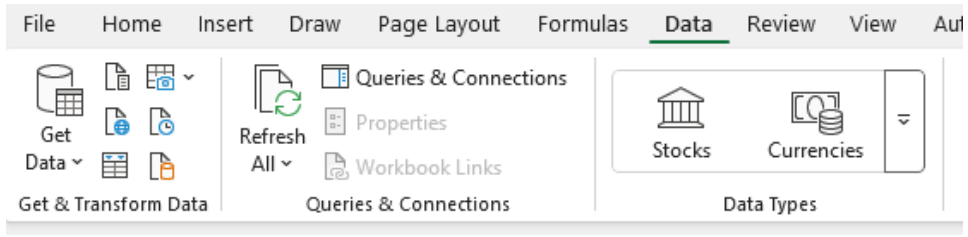


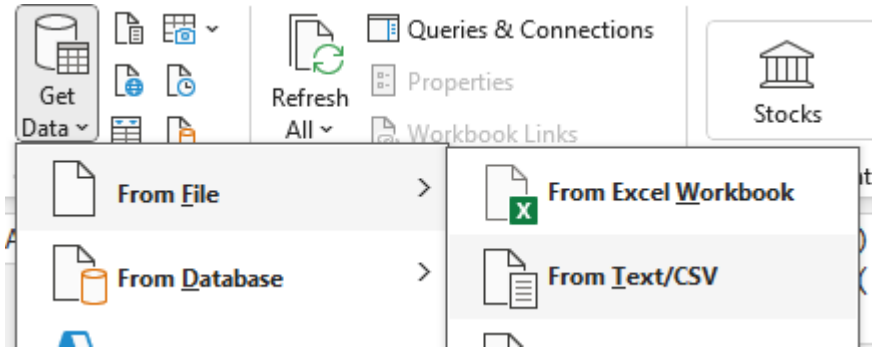
You do have to install the MetaHuman and ARKit plugin to the project. I did find one problem with it, that maybe, I could not figure out. With the multiple facial capture .csv I could only assign one to the ARKit Face Subject on the MetaHuman, which meant, for me at least, that I would only be able to have one set of data controlling the facial rig. So, with the assistance of ChatGPT, I set about combining all facial capture into one .csv file for each character...which follows.

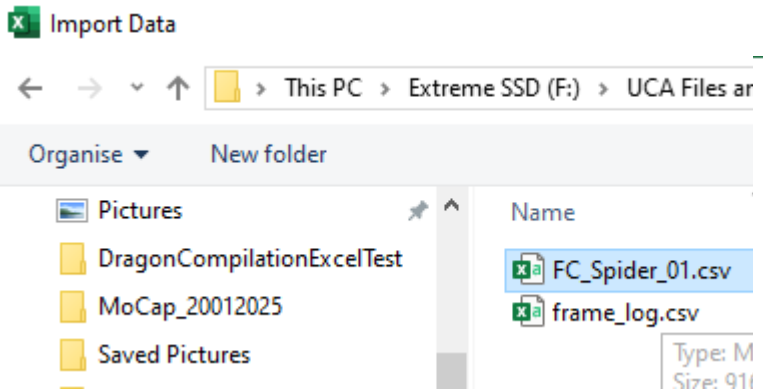


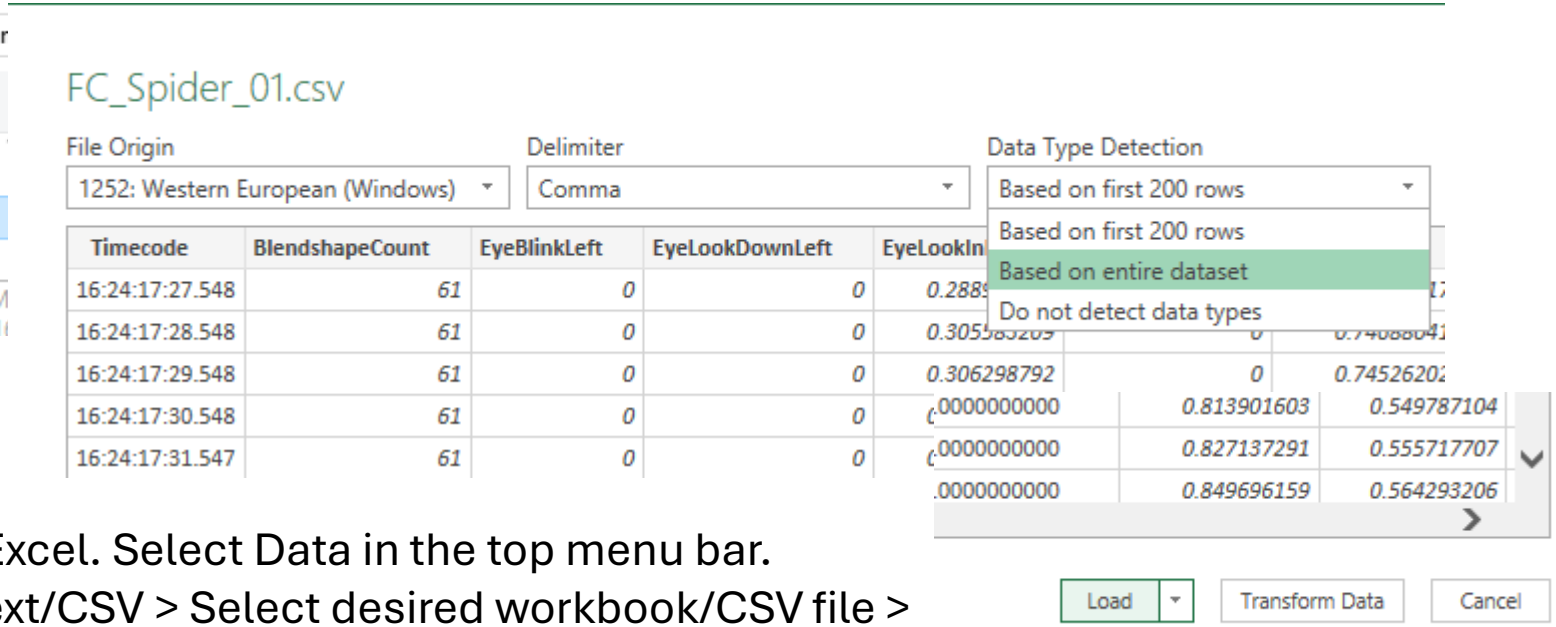


Getting data from different Excel Workbooks/CSV files



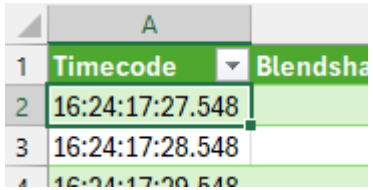


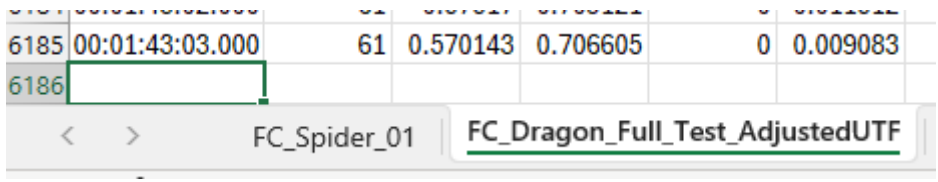


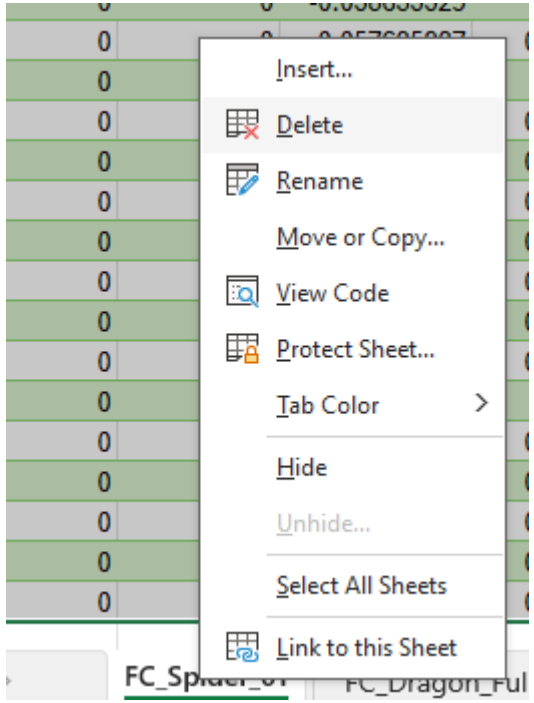


In your primary (target) Workbook in Excel. Select Data in the top menu bar.
Select Get Data > From File > From Text/CSV > Select desired workbook/CSV file >
Data Type Detection : Based on entire dataset > Load

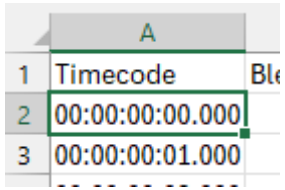






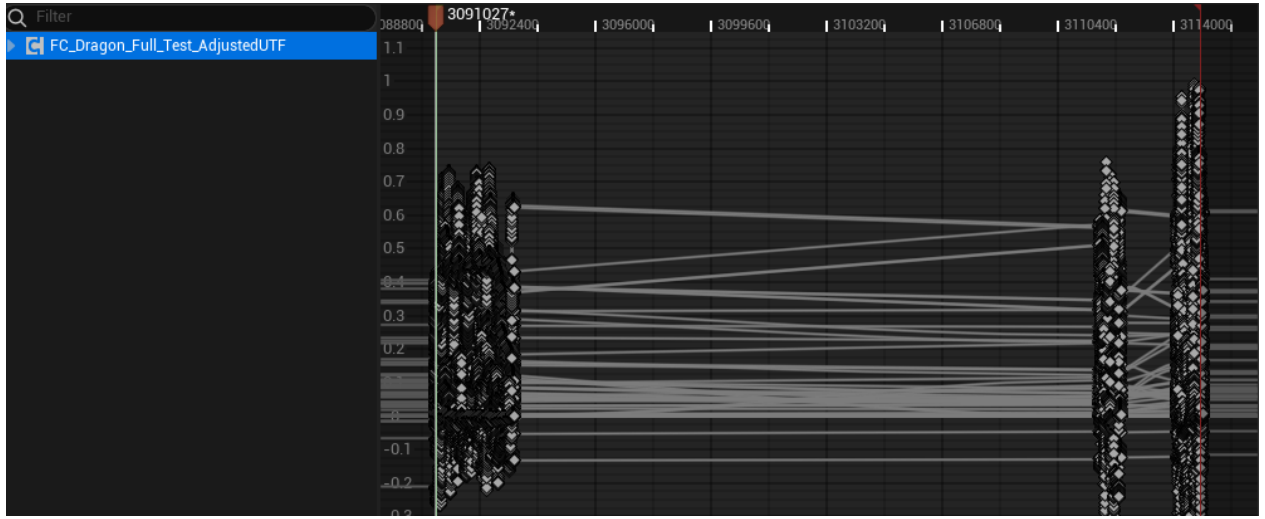


Get Data adds selected Worksheet to current Workbook.
In the donor (imported) Worksheet, Select 1st Data field, Ctrl + A (selects ALL data), Ctrl + C (copies ALL selected data), go to the target Worksheet and select field that you want to paste the data to, then Ctrl + V to paste the copied data.
Select the Worksheet you just copied from and delete it. Make sure there are no blank rows in the data (Right mouse click the row number and select delete). Then in the first data field under Timecode (A2) double click it and enter the formula below, this will reset the Timecode to a sequential dataset for 60 frames per second that UE5 will recognize as a LiveLink .csv file and automatically import it with no issues, fixing any "flat" data you may see without a sequential Timecode.

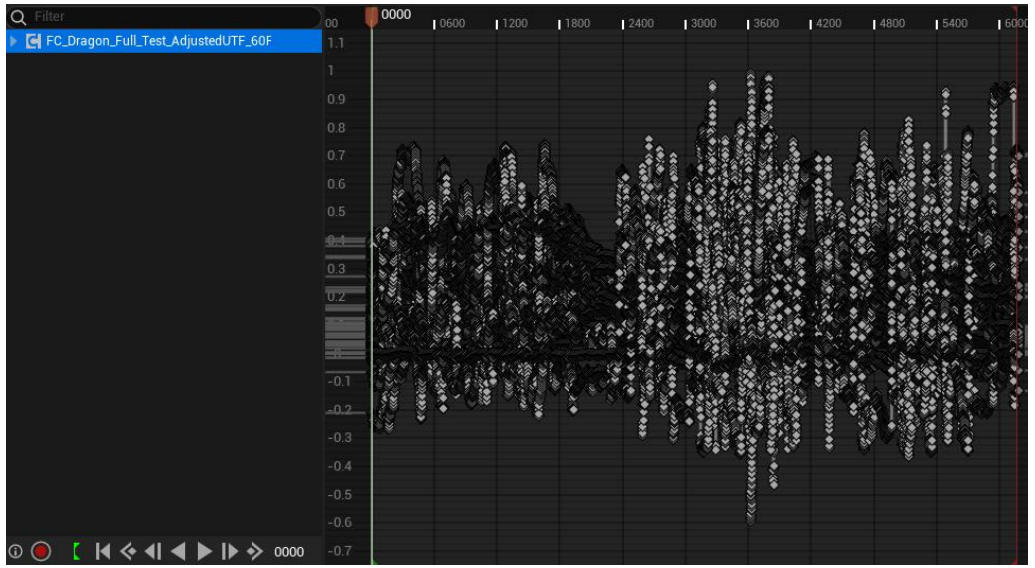


=TEXT(INT((ROW()-2)/216000), "00") & ":" & TEXT(INT(MOD((ROW()-2)/3600,60)), "00") & ":" & TEXT(INT(MOD((ROW()-2)/60,60)), "00") & ":" & TEXT(MOD(ROW()-2,60), "00") & ".000"

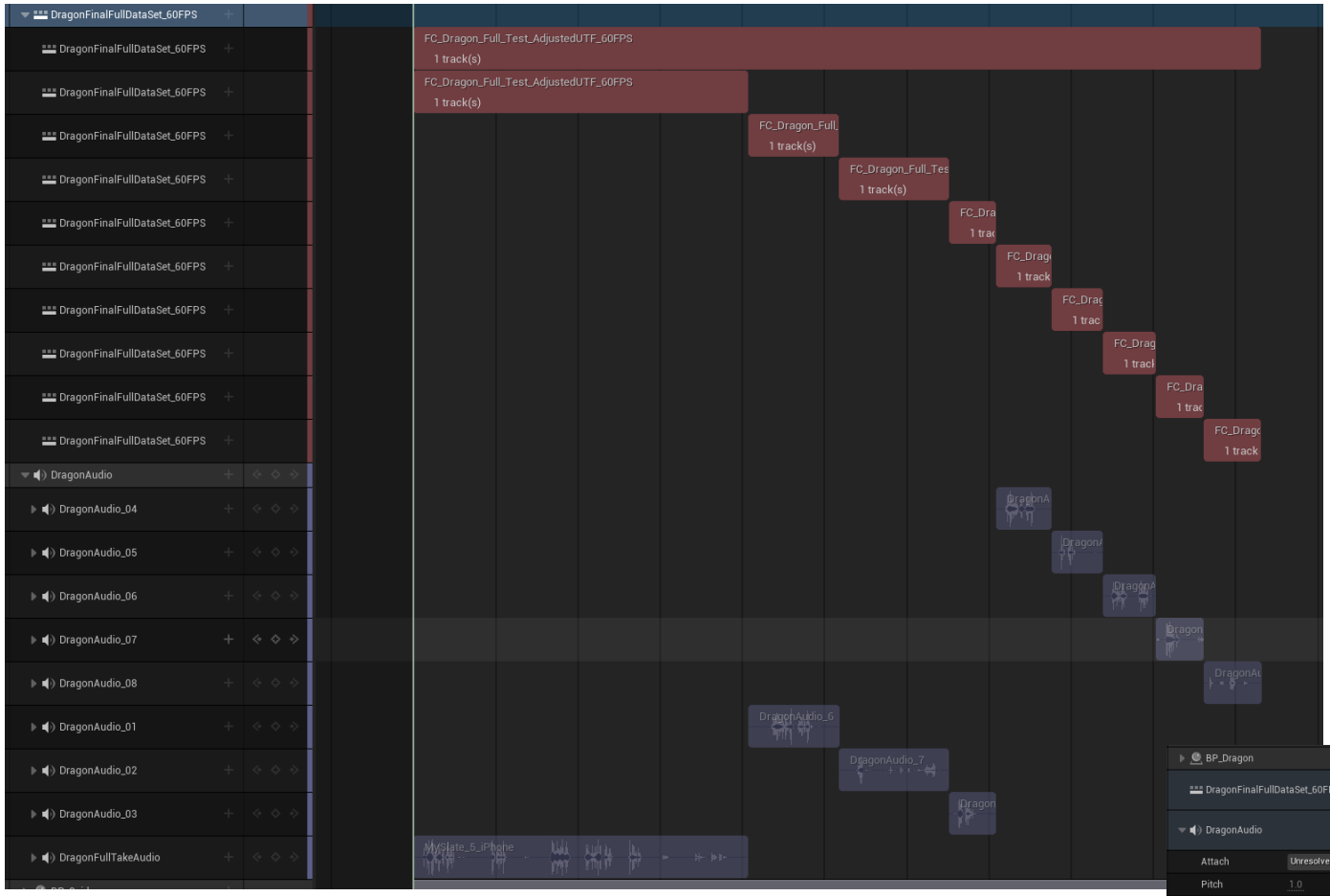
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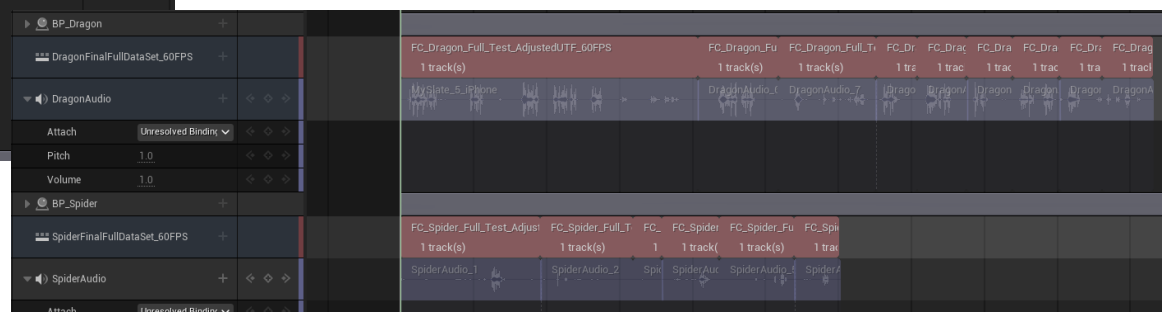


Non-sequential Timecode resulted
In “flat” data



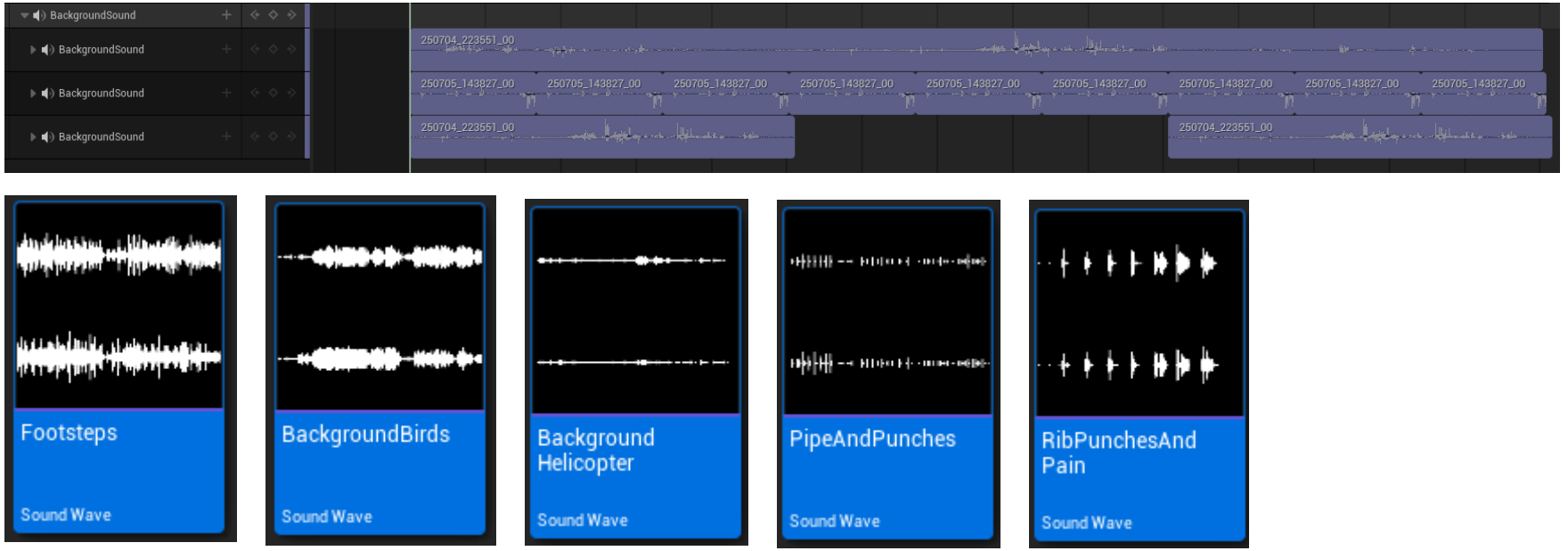
Using the above formula fixed the Timecode
Resulting in a fully populated data set at 60 fps.



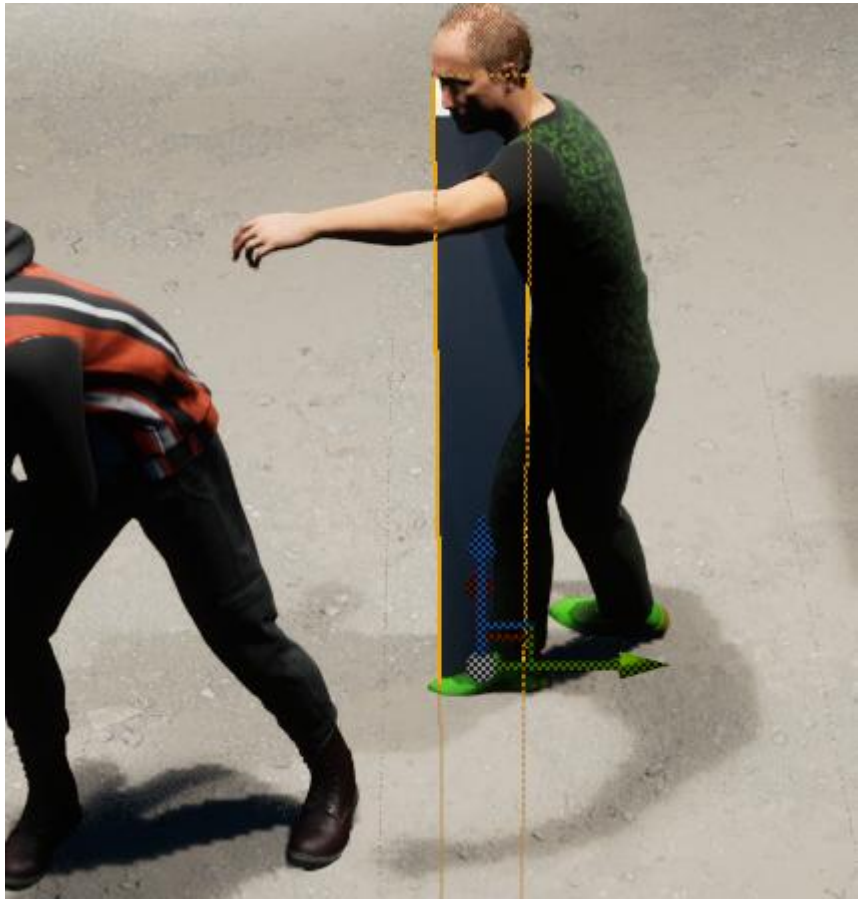


After creating the two full sets of data I then separated them corresponding to the dialogue audio clips order and the number of frames for each clip. I then put them all back into one subsequence and audio track each to minimise space taken up in the animation sequence.

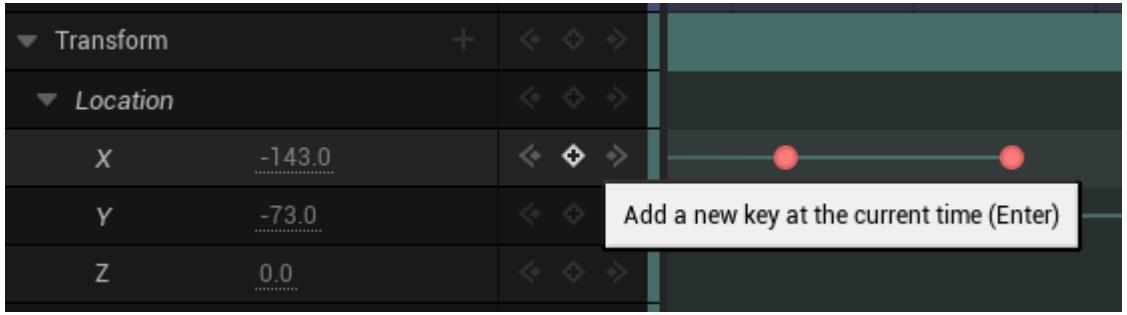


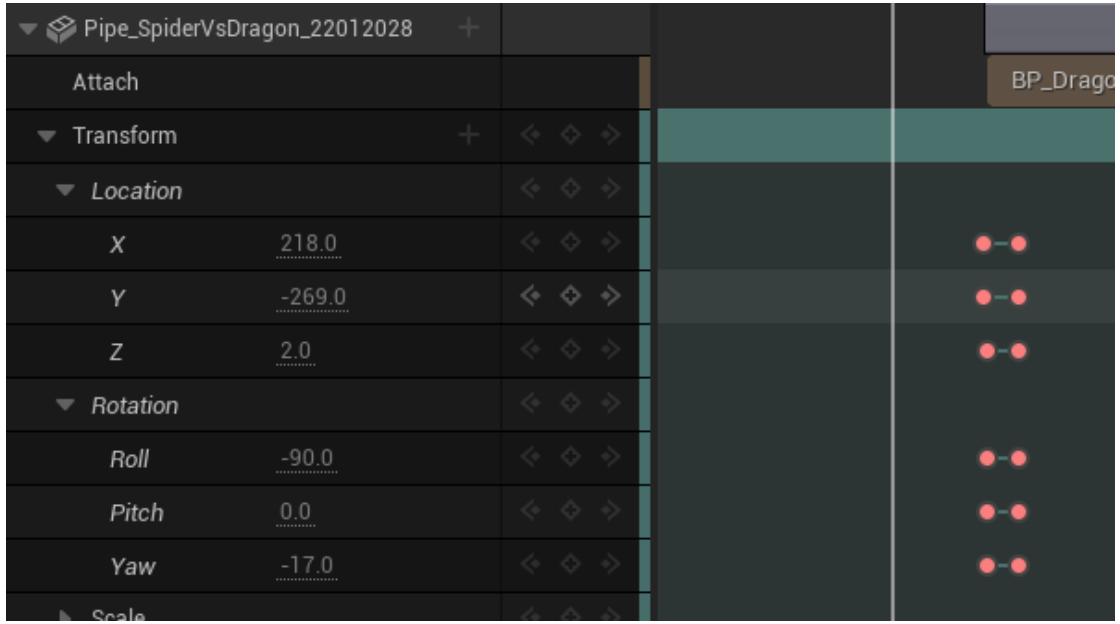


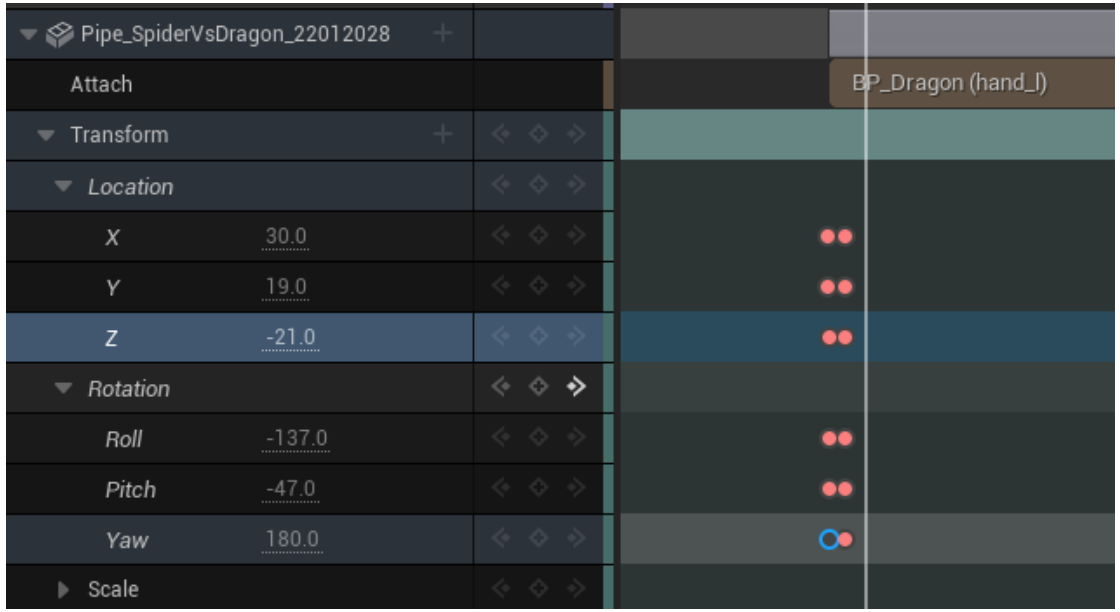
I used my Philips dictaphone to record stereophonic .wav sound files for the sequence. I got lucky with the helicopter for the Background Helicopter recording, the Footsteps are me walking in a field in dried mud tractor tracks next to where I live, it is quite windy where I live so that unfortunately made it into the footsteps recording. Pipe sound is me swinging a wooden bokken (practice katana) and the punch sounds are me hitting myself in the arm, stomach and ribs. Painless... for the most part.



Because some of the animations needed adjustments in Transformation (XYZ) I used a stretched cube as a position marker for the Spider character to aid in more accurate adjustments adding keyframes where necessary.





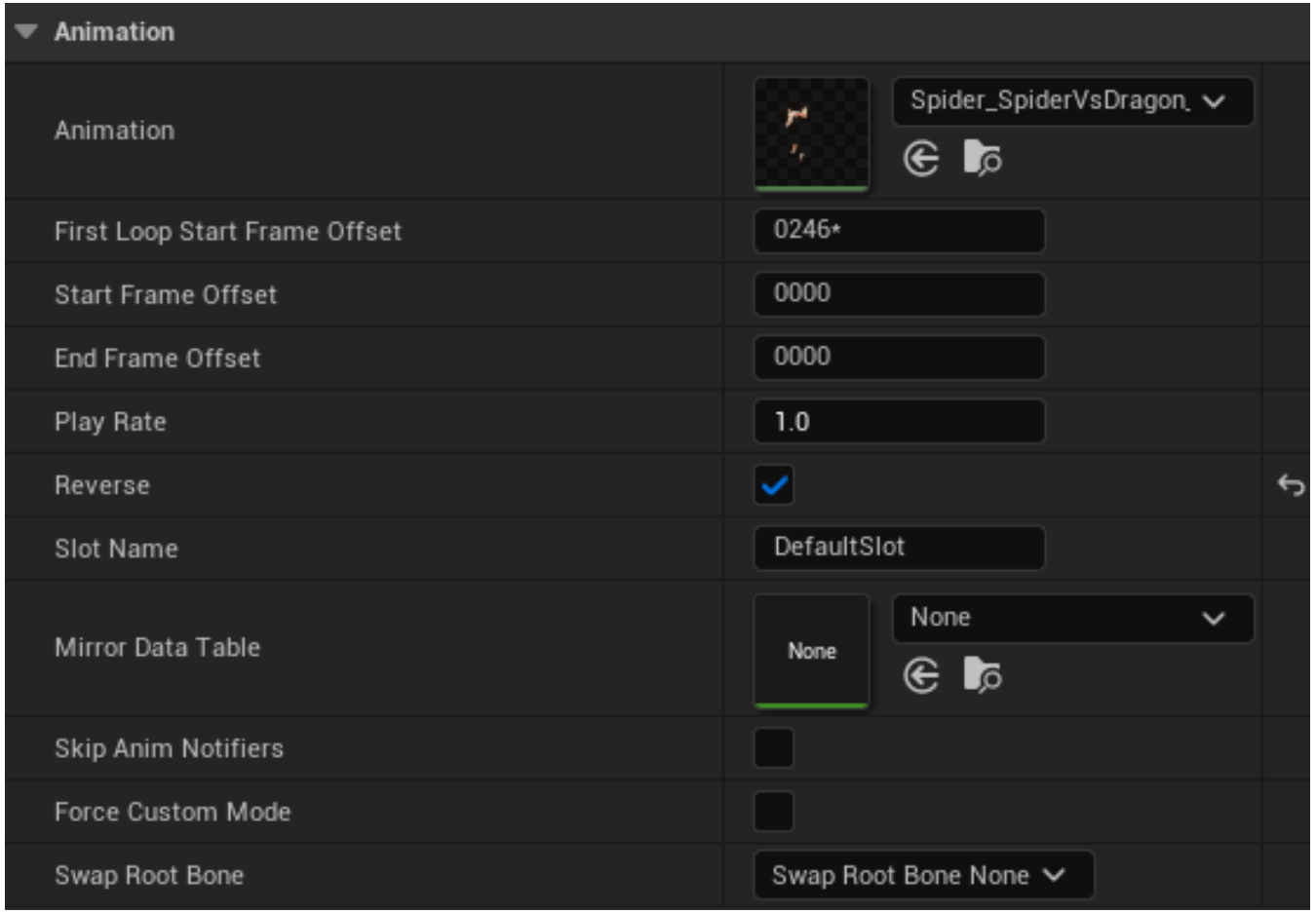


Adjusting Location and Rotation was particularly challenging for the pipe. Originally there was an animation for it but adjusting the Transform to match the Dragon character that was wielding it was difficult so I decided to add a static mesh for the pipe and manipulate that instead, attaching it to the relevant hand in correspondence with the previously captured Pipe animation sequence. Doing it this way also meant I did not have to create another combined data .csv file for the pipe as there were multiple takes for it.



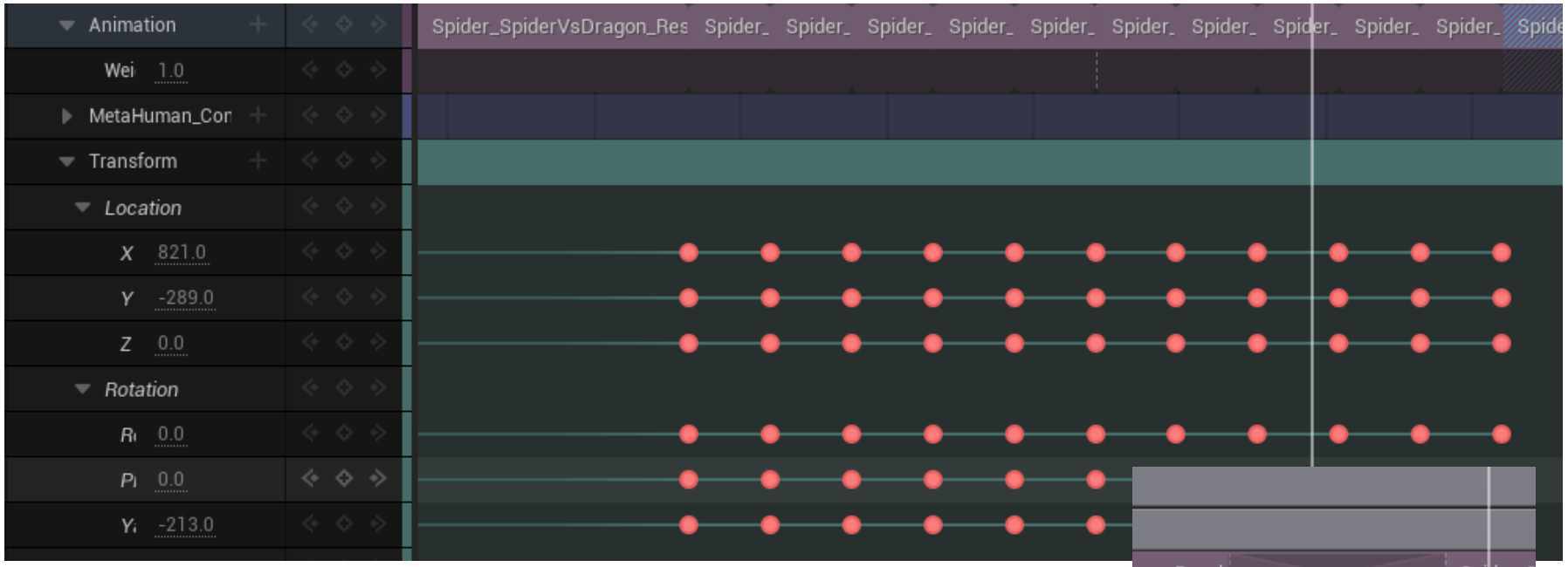
I placed the cameras on one side of the scene. Two for over the shoulder medium close-up shots with 85mm zoom, one wider angle full body with 17mm zoom for side shots to get both characters.

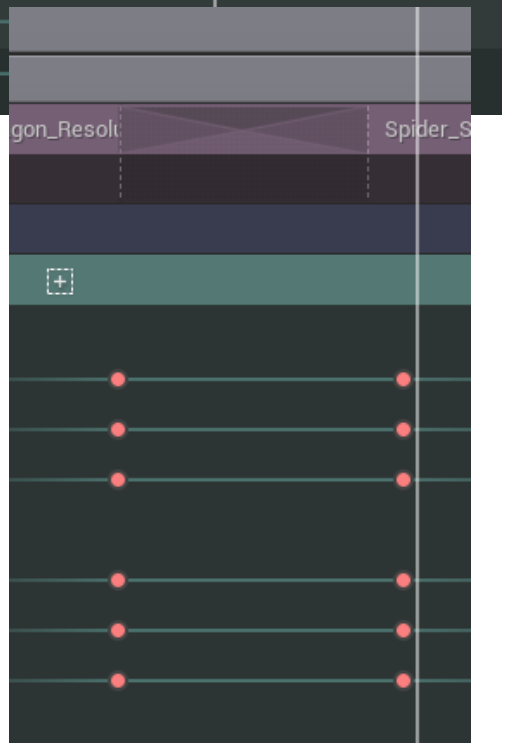




For the kneeing sequence, I copied one kneeing animation and reversed it while blending it to the end of the animation I copied from. I added a second and final kneeing animation to finish the kneeing sequence.

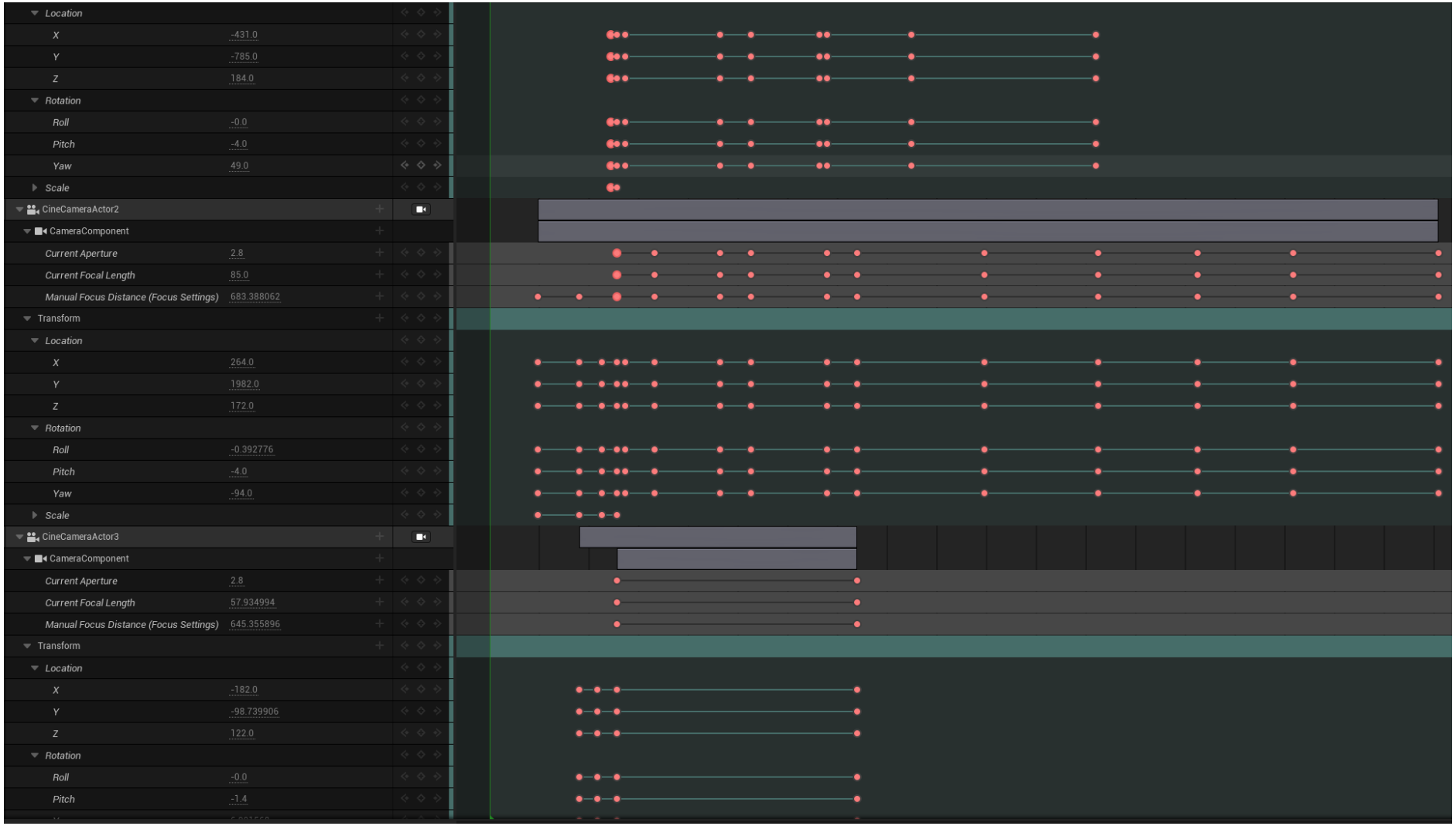




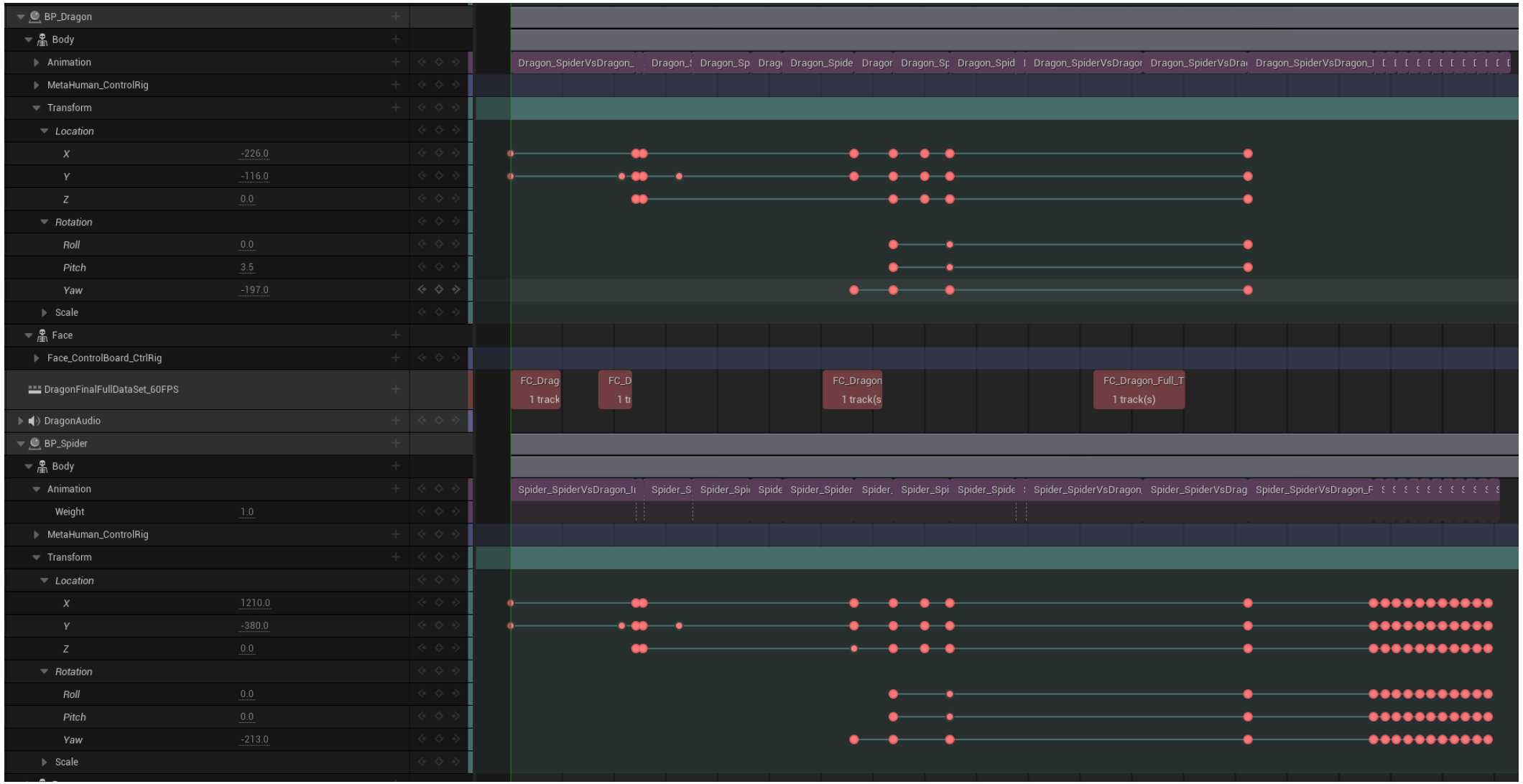


For the final “walk-out” sequence of the Spider character, I used the same cube position marker technique from earlier with the only walk cycle I could find and repeating it with tweaks to location and rotation to match up the end of one cycle with the beginning of the next. It is not perfect, but it was the best I could do. I used a very small blend space between each cycle and adjusted the transformations at either end of the blend space.

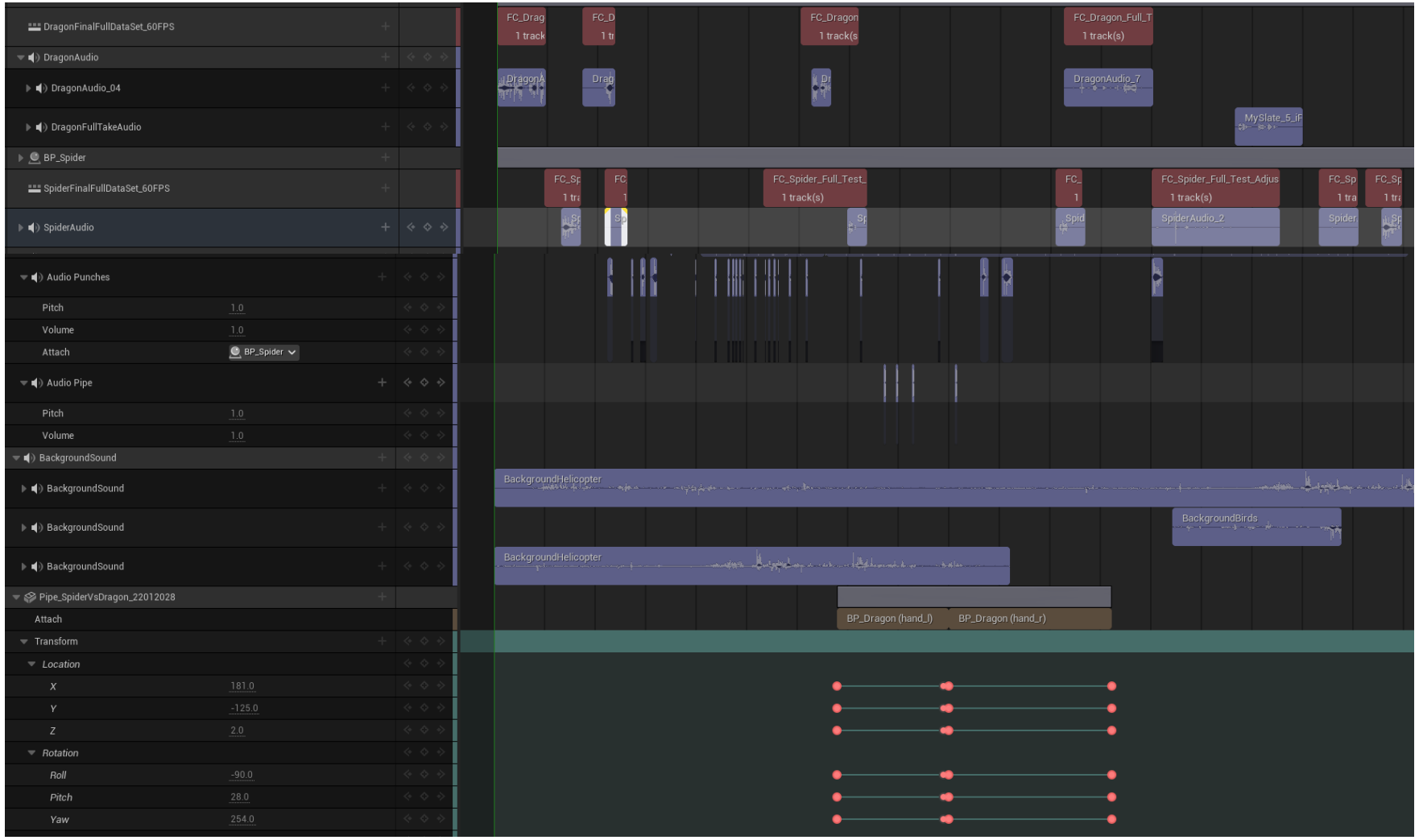
Camera Transforms



Animation Transforms



Facial Capture, Audio Placement and Pipe Attachments/Transforms



Reflection

What I would do differently next time, if there is a next time. For the body motion capture I would make sure that there were start/finish markers for the performers. I had to make a lot of transformations to try and get the animations into similar positions in the environment space. I had no idea how the animations would translate into unreal engine, now I do. There is an origin, 000, and everything is positioned from that. Another way to get round this would be to do the entire motion capture sequence in one take, we did have a lot of short takes. And this would be the same for the Livelink face capture also, I did find a solution for that though. I think it would also be smart to have the motion capture performers do an individual walk/run cycle just in case it might be needed at some point in the future when you're trying to piece together all these small takes of animation. One thing I could not do was hide all the changes in takes in the final sequence with camera cuts as this probably would have induced a seizure. When we were calibrating the hands, we only got to 31% this is the reason that as far as I can tell there are no clenched fists in any of the takes, so if hands are an important part of a motion capture, then I think a higher percentage of calibration would be necessary for us to get the desired result. I enjoyed the creative process for this unit (my script was awesome) and learned a lot about animation both inside and outside of unreal engine.